

Solutions and R Programming Examples: Hypothesis Testing

1.Statcrunch

```
time<-c (6.5,4,3.5,0,12,5,12,15,12,0,1,8,0.5,2,7)
```

```
boxplot (time)
```

```
shapiro.test (time)
```

Shapiro-Wilk normality test

data: time

W = 0.91456, p-value = 0.159

```
t.test (time, mu=7)
```

One Sample t-test

data: time

t = -0.85377, df = 14, p-value = 0.4076

alternative hypothesis: true mean is not equal to 7

95 percent confidence interval:

3.136636 8.663364

sample estimates:

mean of x

5.9

```
qt(0.975,14)
```

```
[1] 2.144787
```

```
pt(-0.854, df=14)
```

```
[1] 0.2037403
```

2. The waiting time

```
wt<-c(4.21,5.55,3.02,5.13,4.77,2.34,3.54,3.20,4.50,6.10,0.38,5.12,6.46,6.19,3.79)
```

```
boxplot(wt)
```

```
shapiro.test(wt)
```

```
Shapiro-Wilk normality test
```

```
data: wt
```

```
W = 0.94898, p-value = 0.5085
```

```
t.test(wt, mu=5, alternative="less")
```

```
One Sample t-test
```

```
data: wt
```

```
t = -1.6867, df = 14, p-value = 0.05691
```

```
alternative hypothesis: true mean is less than 5
```

95 percent confidence interval:

-Inf 5.03157

sample estimates:

mean of x

4.286667

3. The strength of the insulators

```
force<- c(1870,1764,1774,1680,1728,1734,1550,1810,1656,1662,1756,1652,1610,1734,1762,1736,1634,1866,  
1784,1820,1522,1744,1696,1788,1592,1688,1662,1810,1866,1752)
```

```
boxplot(force)
```

```
qqnorm(force)
```

```
qqline(force, col = "steelblue")
```

```
shapiro.test(force)
```

Shapiro-Wilk normality test

data: force

W = 0.97293, p-value = 0.622

```
t.test(force,mu=1500, alternative="greater")
```

One Sample t-test

data: force

t = 13.664, df = 29, p-value = 1.816e-14

alternative hypothesis: true mean is greater than 1500

95 percent confidence interval:

1695.62 Inf

sample estimates:

mean of x

1723.4

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